
Nereus: A Large-Scale Underwater Dataset for Fine-Grained Attribute Understanding and Grounded Counting Perception

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Abstract

Underwater imagery records rich ecological information and supports applications such as science communication, fishery monitoring, ecological conservation, and oceanographic research. Beyond general scene understanding, these applications require models to recognize fine-grained biological traits and reason over spatially grounded evidence. However, existing underwater datasets often provide coarse category labels, generic captions, or total-count supervision, which limits their use for organism-level analysis and ecological reasoning. We introduce **Nereus**, a large-scale dataset for fine-grained underwater perception that jointly supports Fine-Grained Object Attribute Understanding and Grounded Counting Perception. Nereus contains 88K images and 2.68M question-answer pairs, including annotations for 136K marine organism objects across 2,690 species and 1,515 genera, as well as 65K point annotations and 67K grounded counting QA pairs. Experiments with representative multimodal models show that Nereus improves downstream adaptation over existing underwater-domain data and provides a foundation for future research on fine-grained marine perception.

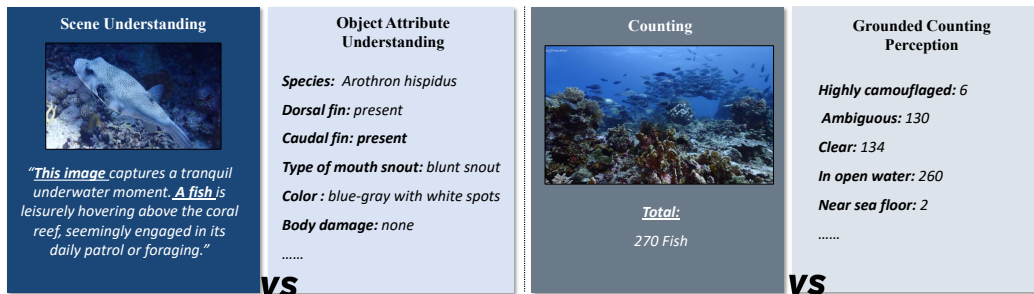


Figure 1: Overview of Nereus. The dataset connects fine-grained organism attributes with grounded counting cues and degradation-aware underwater perception.

1 Introduction

Underwater ecosystems contain rich biodiversity, complex ecological interactions, and valuable natural resources. Large-scale underwater imagery provides an important visual record for science communication, fishery monitoring, ecological conservation, and oceanographic research. Recent progress in computer vision and multimodal large language models (LMMs) makes it increasingly possible to extract ecological knowledge from such imagery and reduce manual observation costs. Unlike in-air visual perception, underwater scenes suffer from severe degradations and domain gaps,

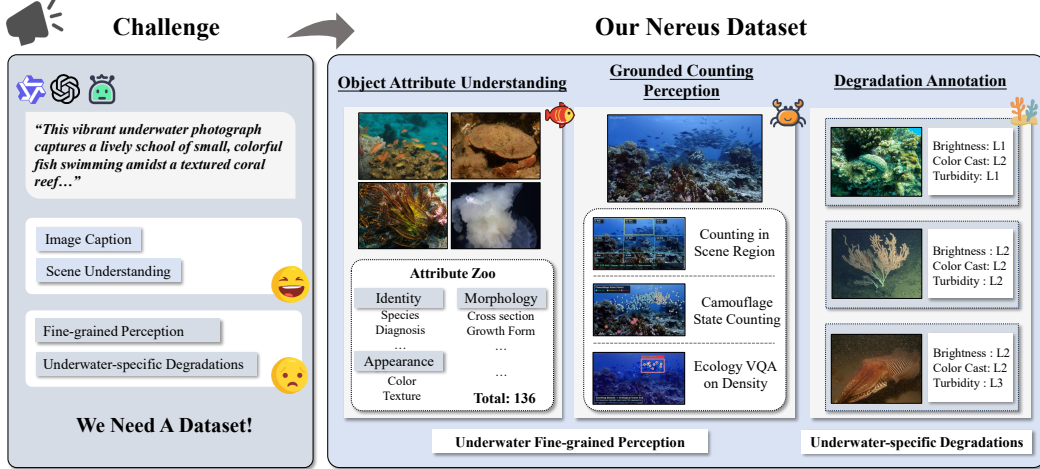


Figure 2: Motivation of Nereus. The figure contrasts application needs in underwater scenarios with the limitations of current underwater understanding resources, motivating a dataset that jointly supports Fine-Grained Object Attribute Understanding, Grounded Counting Perception, and degradation-aware analysis.

such as color cast, scattering, turbidity, blur, low contrast, and occlusion, making fine-grained visual perception highly challenging. while general underwater scene understanding has received increasing attention, fine-grained underwater visual perception remains unexplored.

General underwater vision-language datasets mainly emphasize scene-level descriptions, captions, referring expressions, detection, or broad visual question answering. Taxon-specific datasets provide valuable biological supervision, but are usually restricted to a limited set of organism groups, such as fish or coral. Counting datasets provide point annotations or numerical supervision, yet they usually treat counting as an isolated task of estimating a total number rather than reasoning over semantic regions, visibility conditions, density patterns, or ecological context. As illustrated in Figure 2, this leaves a gap between existing underwater resources and application-oriented underwater perception: current resources are relatively effective for image captioning and coarse scene understanding, but provide limited support for fine-grained visual perception including object-level biological attributes, grounded counting perception, and underwater-specific degradation analysis.

Table 1: Comparison between Nereus and existing underwater understanding datasets. Nereus contains 88K images, 2.68M Question Answer pairs, 136K objects, and 65K point annotations, and mainly supports two tasks highly related to marine downstream applications: Fine-Grained Object Attribute Understanding and Grounded Counting Perception.

Datasets	Venue	Img	Annotation Modality			Supported tasks				
			Object-level Attr. Annotation	QA Pair	Instance Desc.	Detection	VQA	Caption	Fine-Grained Attr. Understanding	Counting Perc.
MarineInst20M	ECCV 2024	2.42M	×	×	✓	✓	×	✓	×	×
NautData	NeurIPS 2025	158K	×	✓	✓	✓	✓	✓	×	×
Fish-Vista	CVPR 2025	69.3K	✓	×	×	×	×	×	✓	×
CoralVQA	NeurIPS 2025	12.8K	✓	✓	×	×	✓	×	✓	×
UWBench	arXiv 2025	15.0K	×	✓	✓	×	✓	✓	×	×
IOCFish5K	CVPR 2023	5.6K	×	×	×	×	×	×	×	✓
Nereus (Ours)		88K	✓	✓	✓	✓	✓	✓	✓	✓

Table 1 further summarizes this gap from the perspectives of data scale, annotation modality, and supported tasks. Existing datasets differ in their strengths: MarineInst20M [1] provides large-scale marine instance descriptions, NautData [2] and UWBench [3] support general underwater vision-language understanding, Fish-Vista [4] and CoralVQA [5] contribute fine-grained supervision for specific taxonomic groups, and IOCFish5K [6] provides point-level supervision for underwater fish counting. However, these resources do not jointly provide structured object-level attribute annotations, Question Answer supervision, descriptions, detection-related localization, captioning support, fine-grained attribute understanding, and counting perception within a unified dataset. In particular,

existing counting resources rarely transform point annotations into region-aware, camouflage-aware, and density-grounded instruction data, which limits their use for fishery monitoring and ecological reasoning.

In this work, we focus on four underwater applications: Science Communication, Fishery Monitoring, Ecological Conservation, and Oceanographic Research. These scenarios motivate **Fine-Grained Underwater Perception**, which requires object-level biological attributes, spatially grounded counting evidence, and degradation-aware interpretation. We introduce **Nereus**, a large-scale underwater dataset organized around two task families: **Fine-Grained Object Attribute Understanding**, which provides cross-taxon object-level supervision over taxonomy, morphology, appearance, condition, pose, visibility, and organism-environment interaction; and **Grounded Counting Perception**, which extends counting from numerical estimation to region-aware, camouflage-aware, and density-grounded reasoning. All images are additionally annotated with luminance, color cast, and turbidity-related degradation levels.

Overall, Nereus contains 88,555 images and 2.68M Question Answer pairs, including 136K annotated marine organism objects across 2,690 species and 1,515 genera, 136 attribute dimensions, 65K point annotations, and 67K grounded counting QA pairs. We further construct downstream evaluations and benchmark representative multimodal models, including Qwen, InternVL, LLaVA, and Gemini. Experiments show that fine-tuning on Nereus consistently improves fine-grained underwater perception over zero-shot baselines and models adapted with existing marine-domain datasets, while the benchmark reveals remaining challenges in subtle attribute recognition, region-grounded counting, camouflage reasoning, and density-based ecological interpretation. Our contributions are summarized as follows:

- We introduce **Fine-Grained Underwater Perception**, an application-oriented multimodal problem that goes beyond general underwater scene understanding. We define two core capabilities: **Fine-Grained Object Attribute Understanding** and **Grounded Counting Perception**, covering science communication, fishery monitoring, ecological conservation, and oceanographic research.
- We build **Nereus**, a large-scale underwater dataset with finer-grained biological attributes and grounded counting supervision. Nereus contains 88K images, 2.68M question-answer pairs, 136K annotated objects, 136 attribute dimensions, 2,690 species, 1,515 genera, and 65K point annotations. Its annotations are further verified through LMM-assisted screening and human review.
- We provide a comprehensive benchmark with representative open-source and closed-source LMMs, including Qwen, InternVL, LLaVA, and Gemini. Experiments show that Nereus improves downstream adaptation over prior underwater-domain data; for example, compared with NautData, it raises overall attribute accuracy from 12.06% to 48.91% on LLaVA-OneVision, 16.08% to 48.10% on InternVL2.5, and 17.09% to 41.55% on Qwen2.5-VL.

2 Related Work

2.1 General underwater scene understanding

Large multimodal models have advanced captioning, VQA, grounding, referring expressions, and multimodal reasoning [7–11], but underwater images introduce domain-specific degradations such as color cast, scattering, turbidity, blur, low contrast, and occlusion [12–15]. Underwater scene resources such as SUIM [16] and recent marine multimodal datasets, including MarineInst20M [1], NautData [2], UW Bench [3], CoralVQA [5], Fish-Vista [4], and MarineGPT [17], support segmentation, instance descriptions, instruction following, captioning, grounding, VQA, or taxon-specific understanding. These works are valuable, but they mainly focus on scene-level understanding, benchmark evaluation, instance description, or specific organism groups, leaving cross-taxon biological attributes and grounded counting perception under-supported.

2.2 Fine-grained underwater visual perception

Fine-grained underwater perception requires models to answer questions about subtle biological and spatial evidence, such as visible organism traits, densest regions, camouflage states, and habitat-

related density patterns. General fine-grained resources such as CUB-200-2011 [18] and EagleVision [19] show the value of structured object attributes, while biological databases such as FishBase [20], WoRMS [21], and coral trait databases [22] provide rich trait knowledge. However, these resources are not directly aligned with underwater visual evidence. Existing underwater datasets provide only partial support through fish traits, coral VQA [5], or general marine descriptions. Nereus complements them by connecting object-level biological attributes with region-aware counting, camouflage-aware counting, and density-hotspot reasoning.

3 Nereus Overview

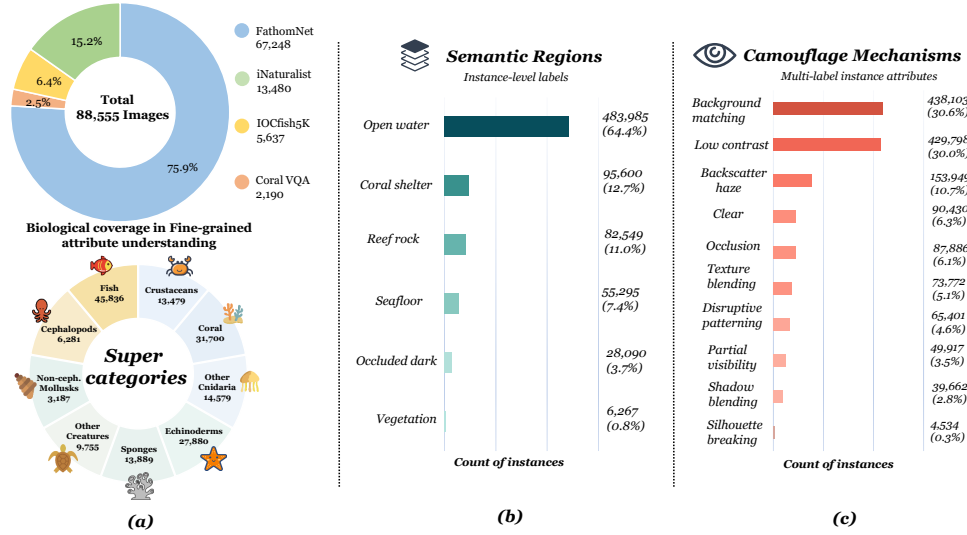


Figure 3: Overview statistics of Nereus. The figure summarizes the data resources used to construct Nereus, the biological coverage and taxonomic diversity of the Fine-Grained Object Attribute Understanding subset, semantic-region statistics for grounded counting, and camouflage mechanism statistics.

3.1 Overview

Data Curation. Nereus is a large-scale underwater dataset for Fine-Grained Attribute Understanding and Grounded Counting Perception. It contains 88,555 images in total and is split into training and test sets, with 86,376 images for training and 2,179 images with 16,744 QA pairs for model evaluation. We construct the test set by sampling according to the distribution of super categories and include species absent from the training set, allowing evaluation of cross-species generalization. We collect and integrate data from four sources: FathomNet, CoralVQA, iNaturalist, and IOCfish5K [23, 5, 24, 6]. FathomNet serves as the primary source for object-level marine organism annotations, CoralVQA and iNaturalist complement coral, sponge, and fish data, and IOCfish5K provides point annotations for counting-oriented perception. The number of samples collected from each dataset is summarized in Figure 3. License information for the data sources and our released annotations is provided in Appendix A.3.

Statistics for Fine-Grained Object Attribute Understanding. Following the biological distribution of FathomNet, we organize marine organisms into nine major groups: Fish, Coral, Cephalopods, Crustaceans, Echinoderms, Non-Cephalopod Mollusks, Other Cnidarians, Sponges, and Other Creatures [23]. Nereus contains object annotations across these groups, and the object-count composition is shown in Figure 3. To better support downstream applications, we define a 136-dimensional attribute set by consulting biological resources such as FishBase, Coral Trait Database, and FishVista [20, 22, 4], as well as by analyzing downstream application requirements. The attribute set covers five categories: identity, structural morphology, appearance and surface, condition and environment, and pose, visibility, and interaction, with an overview of the attribute taxonomy provided in Appendix Figure 7.

125 **Statistics for Grounded Counting Perception.** Nereus supports three counting perception tasks:
 126 region-aware counting, camouflage-state counting, and density reasoning. To support these tasks,
 127 we define underwater scene types and instance-level camouflage states, so that counting supervision
 128 is connected with spatial context and visibility conditions. The proportions of different scenes and
 129 camouflage states are shown in Figure 3.

130 3.2 Annotation Quality and Accuracy

131 Human participation is central to our annotation quality control. We first design a 10-point rating
 132 scale and use GPT-4o to score a small randomly sampled subset of Nereus annotations [25]. We
 133 recruit 3 master’s students in marine biology to manually proofread and calibrate these results, and
 134 finally establish a reliable 10-point scale for LLM-assisted annotation quality assessment [26]. The
 135 calibrated 10-point scale, its four-level human-verification mapping, and the semantic-similarity
 136 scale are provided in Appendix Tables 5–7; these criteria guide test-set scoring and expert arbitration
 137 to ensure annotation accuracy.

138 **Training-set evaluation.** We randomly sample 200 images from the training set and evaluate them
 139 with a four-level quality scale. The training annotations obtain an average score of 9.26, supporting
 140 the reliability of Nereus annotations.

141 **Test-set evaluation.** Using the 10-point scale, we score all annotations in the Nereus test set. The
 142 annotations achieve an average score of 9.14, and 94.8% receive scores of 8 or above. We further
 143 submit annotations with scores below 8 to domain experts; in total, 63 objects are manually arbitrated,
 144 and 43 attributes are revised.

145 4 Construction

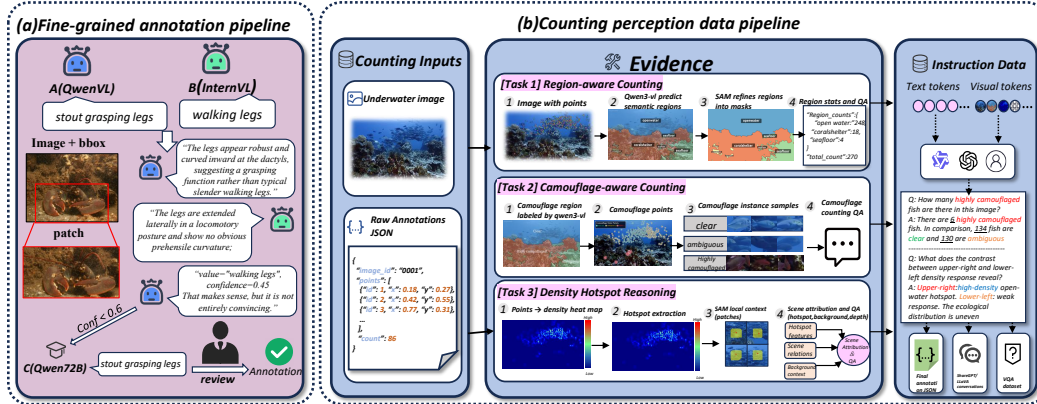


Figure 4: Construction pipeline of Nereus. The pipeline integrates heterogeneous underwater resources. For Fine-Grained Object Attribute Understanding, object-centered crops, localization context, and biological knowledge priors are used to generate structured attribute annotations through multi-model discussion and arbitration. For Grounded Counting Perception, point annotations are converted into semantic-region counts, grid statistics, camouflage labels, density maps, and hotspot evidence, which are further transformed into region-aware, camouflage-aware, and density-reasoning QA pairs.

146 4.1 Fine-Grained Object Attribute Understanding

147 In this section, we describe how Nereus constructs Fine-Grained Object Attribute Understanding
 148 annotations from heterogeneous underwater images. Given the predefined attribute set described
 149 in Section 3, the pipeline focuses on preparing object-centered visual inputs, injecting biological
 150 knowledge priors, and producing reliable annotations through multi-model collaboration and arbitra-
 151 tion.

Object-centered preprocessing. Following EagleVision, we mainly process each annotated organism into an object-centered input so that the annotation model can focus on the target itself [19]. For data without bounding boxes or center-point annotations (iNaturalist), we use GroundingDINO to generate bounding boxes and randomly sample 20% of them for manual review and correction [24, 27]. For data with center-point annotations (CoralVQA), manual review shows that almost all sampled images contain a single dominant subject, which is sufficient for the model to focus on the target organism. After obtaining the target localization, we crop an object patch around the bounding box. We expand the crop with a 15% margin, which preserves nearby visual cues that are useful for interaction-related attributes. More processing details are provided in Appendix A.6. Also, underwater images often suffer from color cast, turbidity, low contrast, and blur, making local object patches insufficient for reliable fine-grained attribute annotation. To address this challenge, we provide the original image together with bounding box coordinates or point annotations as global context, allowing the annotation model to use broader scene semantics when local visual evidence is degraded. We also introduce degradation annotations for all images. Specifically, based on the underwater image quality index (UIQI), we divide underwater degradation into three types and three severity levels, and annotate each image accordingly [28]. Detailed metric definitions, processing procedures, and level partitioning criteria are provided in Appendix A.8.

Knowledge graph construction. To improve annotation accuracy, we construct a knowledge-graph prior from FishBase, WoRMS, Wikipedia, Coral Trait Database, and other biological databases [20, 21, 29, 22]. The knowledge graph contains trait information and textual descriptions for 2,803 species or genera, covering taxonomy, morphology, appearance, habitat, and ecological traits. During annotation, we retrieve the corresponding entry according to each object’s concept name and inject the prior knowledge into the annotation prompt. An example prior prompt and its final annotation fields are provided in Appendix A.7.

Automated annotation. We conduct pilot annotation on the test set and select the best-performing VLM combination [30] from candidate models: Qwen3-VL-8B and InternVL3-8B [31, 32]. We design a “dual-model discussion with single-model arbitration” annotation pipeline. Specifically, model A (Qwen3-VL-8B) and model B (InternVL3-8B) independently annotate each input based on the patch image, original image with bounding box coordinates, attribute definitions, and knowledge-graph prior. The two models then enter a three-turn discussion. In Turn 1, model A reviews model B’s answers and provides agreement decisions and reasons for each attribute. In Turn 2, model B updates its answers according to model A’s feedback. In Turn 3, model A integrates the full discussion and outputs the final consensus value and agreement confidence for each attribute. During arbitration, attributes with confidence below 0.6 are passed to a larger model, model C (Qwen2.5-VL-72B), for independent judgment and final adjudication [9]. Model C can refer to both models’ answers and the discussion history, but is not constrained by their conclusions and may provide a third answer different from both sides. The full automated annotation process is conducted on four RTX 4090 GPUs and takes 7 days to complete.

4.2 Grounded Counting Perception

We convert IOCFish5K point-supervised counting data into grounded instruction data for three sub-tasks: region-aware counting, camouflage-aware counting, and density hotspot reasoning.

Region-aware counting. Given fish points, we first use Qwen3-VL-8B-Instruct to identify coarse underwater semantic regions [31], including open water, coral shelter, reef or rock structures, seafloor, vegetation, and occluded dark areas. SAM then refines these regions into cleaner masks [33]. We assign fish points to the refined regions and to 3×3 grid cells, producing region counts, total counts, densest areas, and spatial comparison statistics for region-aware QA generation.

Camouflage-aware counting. We use Qwen3-VL-8B-Instruct to describe region-level visibility rather than semantic categories. Each region is labeled as clear, ambiguous, or highly camouflaged, with reasons such as low contrast, background matching, texture blending, or occlusion. These labels are propagated to fish points, yielding point- and image-level camouflage annotations for visibility-aware counting questions.

Density hotspot reasoning. Beyond discrete points, we convert fish points into adaptive Gaussian density maps [34], where high-response areas indicate local aggregation. Hotspots are extracted and

used as SAM prompts to crop local foreground and background context. Qwen3-VL-8B-Instruct then connects hotspot patches and surrounding scenes with structures such as coral reefs, open water, seafloor, or vegetation. Representative density-hotspot and density-response QA cases are shown in Appendix Figures 9 and 10.

Grounded counting caption generation. We merge RGB images, point annotations, region statistics, camouflage labels, density maps, hotspot information, depth cues, and scene attributions into final annotation records. Qwen3.5-9B generates natural-language QA pairs [35], while rule-based templates preserve exact numerical supervision for counts, grid statistics, and visibility summaries. The outputs include annotation JSON files, ShareGPT/LLaVA-style conversations, and VQA datasets, with each answer traceable to evidence such as points, regions, camouflage labels, density hotspots, or local patches.

5 Experiment

Table 2: Evaluation results on Fine-Grained Object Attribute Understanding across five attribute categories. Struct. Morph., Appear. Surf., Eco. Health, and Pose Inter. denote Structural Morphology, Appearance and Surface, Ecological Health, and Pose and Interaction, respectively. Nereus fine-tuning uses the structured attribute set as the supervision signal. Best values are shown in bold within each backbone.

Model	Dataset	Identity	Struct. Morph.		Appear. Surf.		Eco. Health		Pose Inter.		Overall	
		Acc $\uparrow$	Acc $\uparrow$	Sim $\uparrow$	Acc $\uparrow$	Sim $\uparrow$	Acc $\uparrow$	Sim $\uparrow$	Acc $\uparrow$	Sim $\uparrow$	Acc $\uparrow$	Sim $\uparrow$
Gemini 2.0 Flash	Zero-Shot	13.84	18.37	55.60	25.50	48.40	14.62	70.80	17.21	48.40	17.86	57.90
	Zero-Shot	4.68	16.82	40.00	14.78	40.90	4.67	77.60	0.85	49.20	10.48	47.40
	NautData	10.66	15.24	44.00	17.53	46.00	7.21	67.00	0.85	46.20	12.06	49.80
	MarineInst20M	18.26	9.20	45.20	32.77	52.30	1.47	22.10	10.05	31.60	13.56	42.10
LLaVA-OneVision-7B	Nereus (Ours)	56.09	41.95	69.60	49.00	68.20	55.21	83.50	46.85	78.60	48.91	74.10
	Zero-Shot	7.26	13.87	46.90	23.28	53.70	3.54	37.10	0.51	30.10	11.12	45.30
	NautData	13.78	18.26	44.70	23.74	51.10	13.15	37.90	2.21	27.60	16.08	43.90
	MarineInst20M	15.20	7.91	30.30	21.82	44.50	1.07	23.10	2.39	24.20	9.97	33.90
InternVL2.5-8B	Nereus (Ours)	54.87	38.68	69.50	55.59	72.50	49.13	88.70	55.71	76.40	48.10	76.20
	Zero-Shot	9.31	10.42	43.50	26.26	45.50	13.62	71.00	6.64	44.60	13.22	51.00
	NautData	13.69	17.76	50.50	20.83	49.40	22.23	53.50	1.36	26.90	17.09	49.60
	MarineInst20M	20.47	12.33	39.90	32.85	53.80	1.00	30.80	4.60	32.50	14.64	43.10
Qwen2.5-VL-7B	Nereus (Ours)	58.06	30.84	64.00	50.00	67.10	39.92	87.60	34.41	72.10	41.55	73.10

5.1 Implementation Details

Models. We evaluate three open-source VLMs with comparable parameter scales but different architectures: LLaVA-OneVision-7B, Qwen2.5-VL-7B-Instruct, and InternVL2.5-8B [8–10]. For zero-shot baselines, we use their pretrained checkpoints and additionally include Gemini 2.0 Flash [11].

Fine-tuning strategy. All comparison and ablation fine-tuning experiments are conducted on four RTX 4090 GPUs and take 12 days in total. We use LoRA while keeping the visual backbone frozen [36]; the trainable modules are limited to language-side LoRA layers, with backbone-specific LoRA targets following each model implementation. Additional training settings and hyperparameters are provided in Appendix A.2.

Evaluation protocols. We evaluate Nereus on two QA-based task families: Fine-Grained Object Attribute Understanding and Grounded Counting Perception, both designed for downstream marine applications. Each sample contains an underwater visual input, optional task-specific grounding evidence, and a natural-language question. Outputs are scored with task-specific normalization, synonym mapping, structured parsing, and LLM-as-a-judge for free-form ecological or grounding responses. For attribute understanding, models predict object-level biological attributes from the image and target localization across five aspects: identity, structural morphology, appearance and surface, ecological health, and pose and interaction. We report category-level accuracy, overall accuracy, and semantic similarity. For counting perception, we evaluate region-aware counting, camouflage-state counting, and density-hotspot reasoning using Count MAE, Exact Count Accuracy, Acc@5, Acc@10, Densest Cell Accuracy, Camouflage F1, Density F1, and Ecology/Grounding F1. Track A uses RGB images to test numerical counting, densest-cell localization, and camouflage-aware reasoning under zero-shot, prompted zero-shot, count-only SFT, QA+structured-final SFT,

Table 3: Nereus counting and density-evidence benchmark. Results are grouped by evaluation track, and best values within each track are shown in bold.

Track A: RGB counting perception						
Backbone	Setting	MAE ↓	Exact ↑	Acc@5 ↑	Cell ↑	Camou. F1 ↑
Gemini 2.0 Flash	ZS	52.80	9.08	11.12	28.79	18.94
	P-ZS	50.50	8.86	13.29	15.50	17.09
	Full	55.50	9.04	12.68	7.62	12.73
Qwen2.5-VL-7B	ZS	98.33	5.51	5.51	7.62	9.58
	P-ZS	94.34	7.26	7.44	4.61	2.38
	Count	38.57	9.92	16.83	6.11	7.41
	QAS	31.98	10.01	18.51	49.16	52.85
	Full	36.19	10.36	16.56	47.39	54.06
LLaVA-OneVision-7B	ZS	113.61	4.63	4.63	0.00	10.26
	P-ZS	104.38	6.24	6.24	0.18	13.33
	Count	79.47	7.26	8.77	0.09	16.90
	QAS	52.81	8.06	12.84	32.68	28.91
	Full	52.27	8.06	12.22	36.67	35.57
InternVL2.5-8B	ZS	87.38	2.22	2.22	22.85	21.26
	P-ZS	167.55	0.00	1.16	20.73	18.98
	Count	31.72	10.01	19.04	20.81	14.72
	QAS	25.99	10.45	20.46	52.70	50.02
	Full	30.48	9.74	18.95	51.90	52.59

Track B: RGB+heatmap density evidence				
Backbone	Setting	Hotspot ↑	Density F1 ↑	Ground. F1 ↑
Gemini 2.0 Flash	ZS	44.88	18.35	16.48
	Full	41.91	18.22	16.15
Qwen2.5-VL-7B	ZS	41.75	18.13	16.04
	D-QA	37.39	45.26	37.19
	Full	35.37	45.24	37.14
LLaVA-OneVision-7B	ZS	32.86	18.98	17.34
	D-QA	1.42	28.27	23.90
	Full	47.48	30.32	25.22
InternVL2.5-8B	ZS	42.13	19.80	18.37
	D-QA	37.59	45.26	36.35
	Full	38.70	43.24	35.00

Track C: Nautilus transfer					
Setting	MAE ↓	Exact ↑	Acc@5 ↑	Cell ↑	Camou. F1 ↑
Nautilus Counting SFT	39.96	9.65	16.30	6.11	8.48
Nereus Count	38.57	9.92	16.83	6.11	7.41
Nereus QAS	31.98	10.01	18.51	49.16	52.85
Nereus Full	36.19	10.36	16.56	47.39	54.06

ZS: zero-shot; P-ZS: prompted zero-shot; Count: count-only SFT; QAS: QA+structured-final SFT; D-QA: density QA SFT; Full: full Nereus setting/SFT.

and full Nereus SFT settings. Track B adds density heatmaps for hotspot localization, density interpretation, and ecological grounding under density zero-shot, density-QA SFT, and full Nereus SFT settings. We also fine-tune Qwen2.5-VL-7B on Nautilus counting QA to evaluate external transfer. The evaluated models include Gemini 2.0 Flash, Qwen2.5-VL-7B, LLaVA-OneVision-7B, and InternVL2.5-8B.

5.2 Main Results

We compare Nereus with NautData and MarineInst20M, two representative underwater or marine-domain datasets for multimodal model adaptation. We evaluate Fine-Grained Object Attribute Understanding across five attribute categories and Grounded Counting Perception across region-aware counting, camouflage-state counting, and density reasoning. For fair comparison, we align the training sample scale across datasets and fine-tune LLaVA-OneVision-7B, InternVL2.5-8B, and Qwen2.5-VL-7B, while also reporting zero-shot results for additional baselines.

As shown in Table 2 and Table 3, models fine-tuned on Nereus consistently outperform zero-shot baselines and models trained on other datasets. For attribute understanding, Nereus improves all five categories across the three open-source backbones. Averaged over these backbones, identity achieves the highest strict accuracy, while structural morphology remains most challenging because it depends on subtle local cues such as fin shape, mouth type, coral growth form, limb structure, and surface details, which are easily affected by underwater degradation.

Gemini 2.0 Flash performs better in the zero-shot setting, reflecting stronger general visual-language priors and instruction following. However, after Nereus fine-tuning, open-source models substantially surpass the closed-source zero-shot baseline, indicating that domain-specific supervision is more effective than general multimodal knowledge for fine-grained underwater perception.

Compared with NautData and MarineInst20M, Nereus shows clear gains in application-oriented categories such as ecological health and pose-and-interaction. NautData mainly provides hierarchical underwater instruction-following data, while MarineInst20M focuses on instance-level marine descriptions; both help general scene understanding but provide limited support for structured biological attributes and grounded counting evidence. By linking object-level attributes, degradation-affected visual evidence, and counting cues, Nereus better captures organism condition, environmental stress, behavior, habitat relation, and visibility interaction, which are central to ecological conservation, oceanographic research, and fishery monitoring.

5.3 Ablation Study

We further study how different supervision formats and degradation-aware inputs affect model performance. For Fine-Grained Object Attribute Understanding, we fine-tune Qwen2.5-VL-7B with

three supervision formats at a comparable data scale: single-attribute QA pairs, object-level captions, and structured attribute sets. Table 4 compares these supervision signals under strict accuracy and semantic similarity.

Table 4: Ablation study on different supervisory signals for Fine-Grained Object Attribute Understanding using Qwen2.5-VL-7B. Object-level Caption denotes natural-language object descriptions. Acc is averaged over all attribute questions within each category.

Supervisory Signal	Identity	Struct. Morph.		Appear. Surf.		Eco. Health		Pose Inter.		Overall	
	Acc ↑	Acc ↑	Sim ↑	Acc ↑	Sim ↑	Acc ↑	Sim ↑	Acc ↑	Sim ↑	Acc ↑	Sim ↑
QA Pairs	10.98	33.64	49.80	33.38	52.00	77.90	82.00	52.13	70.80	39.08	55.20
Object-level Caption	20.99	17.18	57.70	27.87	55.50	3.94	64.60	11.93	68.00	16.78	61.60
Structured Attribute Set	58.06	30.84	64.00	50.00	67.10	39.92	87.60	34.41	72.10	41.55	73.10

The results show a useful distinction between exact matching and semantic consistency. QA-pair supervision obtains strong accuracy on ecological-health and pose-interaction attributes, likely because these categories contain more repeated answer patterns and the QA format directly matches the evaluation query. However, its identity accuracy is low, since species prediction has a much larger and more long-tailed label space, and isolated QA pairs provide weaker cross-attribute constraints. In contrast, structured attribute supervision is not the best on every strict-accuracy column, but it achieves the best overall accuracy and consistently higher semantic similarity. This suggests that structured annotations help the model learn relations between visual evidence, species identity, morphology, appearance, and interaction attributes, so even when an exact normalized label is missed, the prediction remains closer in meaning.

Figure 5 shows representative underwater degradation examples in Nereus. The figure remains in the main text to illustrate the visual conditions considered by Nereus, while the detailed degradation-aware setup, full results, and interpretation are provided in Appendix A.9.

For Grounded Counting Perception, we conduct an additional image-only ablation on InternVL2.5-8B to isolate the effect of different counting supervision forms. As shown in Appendix Table 11, count-only supervision mainly improves numerical counting, reducing MAE from 87.38 to 31.72, but brings limited gains in spatial and camouflage reasoning. QA+structured-final supervision further lowers MAE to 25.99 and greatly improves Densest Cell Accuracy and Camouflage F1, showing that region, grid, and visibility annotations help the model learn grounded counting evidence rather than only total counts. The full Nereus setting achieves the best Camouflage F1, indicating that richer supervision is particularly useful for visibility-aware reasoning.

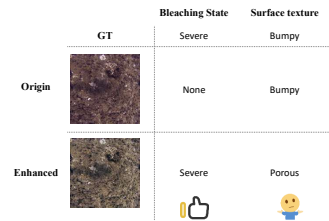


Figure 5: Degradation examples.

6 Conclusion

We introduce Nereus, a large-scale underwater dataset for Fine-Grained Object Attribute Understanding and Grounded Counting Perception. Nereus contains 88K images and 2.68M Question Answer pairs, covering 136K marine organism objects across 2,690 species and 1,515 genera, as well as 65K point annotations for grounded counting. Built around downstream marine applications, Nereus provides fine-grained biological attributes, region- and camouflage-aware counting supervision, density reasoning evidence, and image-level degradation labels. We further benchmark representative open-source and closed-source LMMs, showing that training on Nereus improves fine-grained underwater perception while current models still struggle with subtle attributes, grounding, degradation, and reasoning.

More broadly, Nereus connects organism-level biological understanding with spatial evidence and underwater-specific visual conditions, offering a unified resource for studying marine perception beyond generic captioning or total-count prediction. We expect it to support future work on reliable ecological monitoring, degradation-aware model adaptation, and multimodal reasoning over dense underwater scenes. Its annotations and protocols also support model diagnosis and expert-guided marine analysis. Limitations and future directions are discussed in Appendix A.5.

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429 **Nereus: A Large-Scale Underwater Dataset for**
430 **Fine-Grained Attribute Understanding and**
431 **Grounded Counting Perception**
432 **(Supplementary material)**

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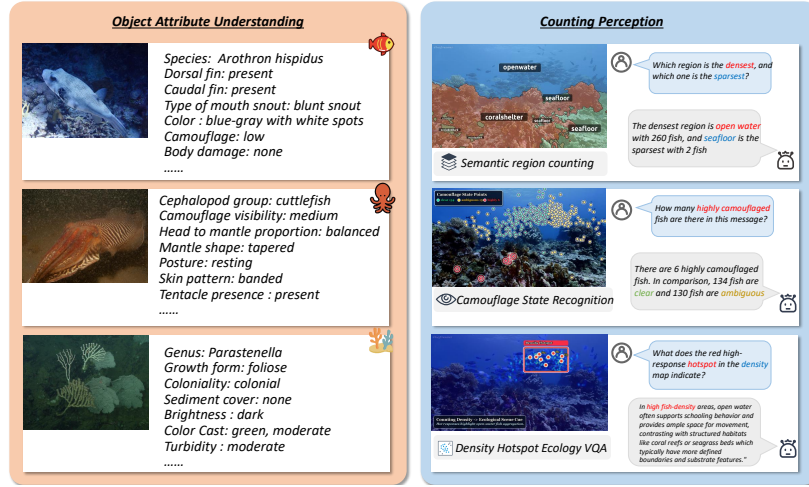


Figure 6: Examples of Nereus applications, covering object-level attribute understanding, region-aware counting, camouflage-aware counting, density-hotspot reasoning, and underwater degradation annotation.

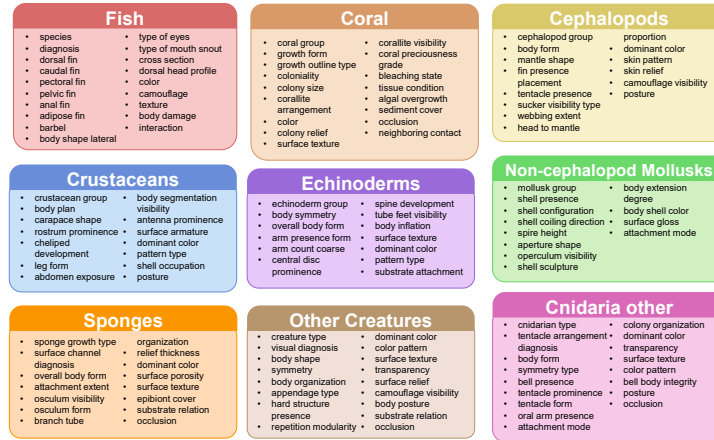


Figure 7: Overview of the Nereus attribute taxonomy. The taxonomy organizes the 136-dimensional structured attribute set into identity, structural morphology, appearance and surface, condition and environment, and pose, visibility, and interaction categories.

451 **A.1 Evaluation scales for annotation quality**

452 We use the following calibrated scales to evaluate annotation quality, support human verification,
453 and compute semantic similarity for free-form underwater attribute responses.

Table 5: 10-point scale for automated annotation quality assessment.

Score	Criterion and descriptive requirement
10	Perfectly accurate, complete, internally consistent, and highly aligned with image evidence.
9	Accurate overall, with only trivial imperfections that do not affect the core understanding.
8	Mostly accurate; minor issues, such as slight terminology bias, but no critical errors.
7	Mostly correct; some secondary attributes show deviations, but core understanding is intact.
6	Partially correct; the core category is right, but notable errors or omissions affect fine-grained details.
5	Borderline usable; only basic attributes are trustworthy, with frequent deviations.
4	Poor; multiple obvious errors, substantial omissions, and few correct attributes remain.
3	Bad; major attributes are wrong and misleading overall, with only incidental correct statements.
2	Very bad; the vast majority of data is wrong, missing, or contradictory.
1	Completely wrong or irrelevant; contradicts image evidence with no redeeming accuracy.

Table 6: Four-level scale for manual annotation accuracy and mapping.

Grade	Mapping score	Qualitative criterion
A	10	Accurate: reflects visual features well, is consistent with the image, and has only minor flaws.
B	7	Acceptable: generally correct, captures main features, and supports basic understanding.
C	4	Deficient: partially correct but contains obvious errors and has limited reference value.
D	1	Failed: inconsistent with the image, with severe errors or internal conflicts.

Table 7: 100-point scale for SIM-based semantic similarity assessment.

Score range	Semantic alignment criterion
90–100	Semantically equivalent, synonymous, or fully correct.
70–89	Correct core meaning with minor differences in granularity or phrasing.
50–69	Correct direction or intent but imprecise or slightly vague.
30–49	Related to the topic but factually or logically clearly wrong.
10–29	Nearly irrelevant; contains only marginal keywords related to the domain.
0–9	Completely wrong, empty, or nonsensical.

A.2 Training details

We use LoRA-based supervised fine-tuning for all open-source backbones [36]. The visual backbone remains frozen across models. For Qwen2.5-VL-7B, training is conducted with LLaMA-Factory [37], and LoRA adapters are applied to language-side attention and MLP layers. For InternVL2.5-8B, we use the native InternVL training framework and train only LLM LoRA parameters, while freezing the LLM base weights, the cross-modal MLP projector, and the vision backbone. For LLaVA-OneVision-7B, we use HuggingFace Trainer with PEFT [38], freeze the vision tower and multimodal projector, and train language-model LoRA parameters. The exact learning rates, LoRA ranks, batch sizes, sequence lengths, and image-resolution constraints follow the corresponding backbone and task settings. Across experiments, we use BF16 precision and train on RTX 4090 GPUs.

A.3 License details

The annotations in FathomNet are released under the CC-BY-ND 4.0 license, while the images are released under the CC-BY-NC-ND 4.0 license [23]. Following these license requirements, we do

not directly package or redistribute the original FathomNet images in our dataset release. Instead, we retain the FathomNet image URLs in the annotation files so that users can download the images themselves. CoralVQA and iNaturalist data follow their respective open licenses [5, 24]. The attribute annotations constructed in this work are released under the CC-BY 4.0 license.

A.4 Broader impacts

Nereus is intended to support research on underwater visual understanding, marine biodiversity analysis, ecological conservation, fishery monitoring, and science communication. By providing fine-grained organism attributes, grounded counting evidence, and degradation-aware annotations, the dataset may reduce the cost of manual visual inspection and help researchers study abundance, habitat association, visibility conditions, and organism-level traits at larger scale. These capabilities can benefit ecological assessment, long-term reef and fishery monitoring, educational tools, and assistive systems for marine scientists.

At the same time, models trained on Nereus should be used with care. Fine-grained recognition and counting systems may be unreliable under domain shift, rare taxa, severe turbidity, unusual imaging devices, or regions underrepresented in the source data. Incorrect predictions could mislead ecological interpretation if used without expert review. The dataset may also be misused for intensive resource extraction, intrusive monitoring of sensitive habitats, or disclosure of locations associated with vulnerable species when combined with external metadata. In addition, LMM-assisted annotations and automatic preprocessing can introduce systematic errors, and image-level degradation indicators cannot fully describe local water quality or hidden biological states.

We therefore recommend using Nereus as a research and decision-support resource rather than a replacement for expert ecological assessment. Users should respect the licenses of original image sources, audit model behavior before deployment, report uncertainty in high-stakes settings, and avoid releasing sensitive location information for vulnerable species or habitats. Future releases should continue expanding taxonomic, geographic, depth, and device coverage, and should include broader expert validation where the dataset is used for conservation or policy-relevant decisions.

A.5 Limitations

Nereus is built from available underwater imagery and therefore may still have uneven coverage across rare taxa, geographic regions, depths, and imaging devices. In addition, parts of the annotation pipeline rely on object detection, LMM-assisted labeling, and image-level degradation indicators, which can introduce noise and cannot fully capture local water-quality variation or hidden biological states. Future work will expand expert verification, temporal and video-based observations, and broader real-world deployment evaluation.

A.6 Additional data processing details

For CoralVQA, which only provides point annotations, bounding boxes are unavailable. We therefore retain only images with a single genus label, meaning that all point annotations in the same image belong to the same genus. This ensures that the whole image can serve as a valid annotation unit for that genus. During data cleaning, we correct label spelling errors, such as correcting Porties to Porites and Orcbicella to Orbicella. We also split labels with bleaching prefixes, such as “Bleached Montipora,” into two independent fields: genus and bleaching status.

For iNaturalist data without bounding box annotations, we use GroundingDINO to detect target objects and obtain bounding boxes. We then apply the same patch cropping and quality filtering pipeline as used for FathomNet.

A.7 Knowledge-prior prompt example

During annotation, the concept name of each object is used to retrieve the corresponding knowledge-graph entry, which is then inserted into the model prompt as reference context. The following example shows the prior prompt for *Merluccius productus* and the resulting structured annotation fields.

KG prior prompt.

Knowledge prior for *Merluccius productus* (Fish): habitat: pelagic-neritic; max length: 83.0 cm; family: Merlucciidae; order: Gadiformes. Note: these are reference values; actual observation may differ.

Final annotations.

Field	Value
adipose_fin	absent
anal_fin	absent
barbel	absent
body_damage	none
body_shape_lateral	elongate
camouflage	low
caudal_fin	present
color	silver-gray
cross_section	circular
diagnosis	Silver-gray body with black speckling
dorsal_fin	present
dorsal_head_profile	straight
interaction	[schooling]
pectoral_fin	absent
pelvic_fin	absent
species	<i>Merluccius productus</i>
texture	smooth
type_of_eyes	small-eyed
type_of_mouth_snout	terminal mouth

A.8 Degradation indicator details

To support evaluation experiments on degradation-aware perception, we compute image-level degradation indicators for all 88,555 images in the dataset. Following the underwater image quality assessment method introduced by UIQI, we extract quality-aware features along three degradation dimensions: brightness, color cast, and turbidity [28].

Brightness. We first convert each image to grayscale and compute its normalized mean luminance:

$$l = \frac{1}{|\Omega|} \sum_{x \in \Omega} \frac{I_{\text{gray}}(x)}{255}, \quad l \in [0, 1],$$

where a larger value indicates a brighter image.

Color cast. We convert the image to the CIELAB color space and compute the pixel-wise means m_a, m_b and variances v_a, v_b of the a and b channels. The overall color mean m , color variance v , and color-cast degree d are defined as

$$m = \sqrt{m_a^2 + m_b^2}, \quad v = \sqrt{v_a^2 + v_b^2}, \quad d = \frac{m}{v + \epsilon}.$$

A larger d indicates a more severe color cast. The color direction is determined by the polar angle on the a - b chromaticity plane,

$$\theta = \text{atan2}(m_b, m_a),$$

which is used to categorize the color cast into blue, green, blue-green, yellow, and other types.

Turbidity. Underwater scattering often causes haze, contrast reduction, and loss of fine details. We capture these degradation patterns with five complementary features: (1) the high-frequency sub-band energy E_T obtained from three-level CDF 9/7 wavelet decomposition, which reflects local blur; (2–3) the shape parameter λ and scale parameter k obtained by fitting a generalized Gaussian distribution to mean-subtracted contrast-normalized coefficients, which reflect global haze; (4) the local contrast energy CE computed on the gray, yb, and rg channels; and (5) the Jensen-Shannon divergence D_{JS} between the grayscale histogram and the uniform distribution, which reflects global contrast.

To avoid interference from brightness and color cast when estimating turbidity, all five features are computed on preprocessed images. The preprocessing includes brightness normalization, which

rescales the mean luminance to 128, and gray-world-based white balance. The standardized features are then fused by principal component analysis (PCA), and the first principal component is used as the final turbidity score:

$$s_{\text{tur}} = \text{PC}_1 \left(\text{standardize} \left([E_T, \lambda, k, CE, D_{JS}]^T \right) \right).$$

A.9 Degradation-aware ablation results

We investigate whether degradation priors and image enhancement can help Fine-Grained Object Attribute Understanding. We compare the original-image baseline with two degradation-aware settings: adding degradation type and level as textual priors, and applying DTIUIE enhancement together with the same priors [39]. Table 8 reports the degradation-aware training ablation for Ultra Fine-Grained Object Attribute Understanding using Qwen2.5-VL-7B, with representative degradation examples shown in Figure 5.

Table 8: Ablation study on degradation-aware training strategies for Ultra Fine-Grained Object Attribute Understanding using Qwen2.5-VL-7B.

Setting	Identity	Struct. Morph.		Appear. Surf.		Eco. Health		Pose Inter.		Overall	
	Acc ↑	Acc ↑	Sim ↑	Acc ↑	Sim ↑	Acc ↑	Sim ↑	Acc ↑	Sim ↑	Acc ↑	Sim ↑
Original Img.	58.06	30.84	64.00	50.00	67.10	39.92	87.60	34.41	72.10	41.55	73.10
Degradation Prior + Original Img.	56.30	29.30	66.20	48.16	66.80	40.25	89.50	35.95	76.20	40.51	74.60
Degradation Prior + Enhanced Img.	51.95	25.27	62.30	45.56	66.20	38.18	86.40	36.12	76.40	37.36	71.80

Degradation priors improve semantic similarity across most attribute categories, with clear gains on structural morphology, ecological health, and pose-interaction attributes. This suggests that explicit degradation information can help the model interpret ambiguous visual cues affected by turbidity, color cast, or blur. However, using enhanced images does not further improve performance and instead slightly degrades most accuracy and similarity scores. This indicates that image enhancement for downstream marine understanding should be task-driven rather than optimized only for perceptual visual quality.

B Counting and Density-Evidence Benchmark

This section provides additional details for Table 3. The benchmark is organized into two evaluation tracks. Track A uses only RGB images and evaluates whether a model can estimate fish abundance, localize the densest area, and reason about camouflage or visibility. Track B uses RGB images together with density heatmap evidence and evaluates whether a model can read density evidence, localize hotspots, and produce grounded ecological interpretations. The main paper reports a compact version of these results; here we provide the expanded transfer comparison, result interpretation, and metric definitions.

B.1 Expanded Nautilus Transfer Comparison

Table 9 expands the Nautilus transfer result summarized in Panel C of Table 3. All models are Qwen2.5-VL-7B variants evaluated on the same Track A test split with 1,129 counting samples. The Nautilus row uses Nautilus counting supervision, while the Nereus rows use different subsets of our grounded counting supervision.

Table 9: Expanded transfer comparison on the Track A counting test split. Nereus variants are compared with Nautilus counting SFT under the same Qwen2.5-VL-7B backbone.

Training data / setting	MAE ↓	RMSE ↓	Exact ↑	Acc@5 ↑	Acc@10 ↑	Densest Cell ↑	Camou. F1 ↑	Count
Nautilus counting SFT	39.96	158.50	9.65	16.30	24.53	6.11	8.48	1129
Nereus count-only SFT	38.57	89.59	9.92	16.83	26.66	6.11	7.41	1129
Nereus QA+structured-final SFT	31.98	78.25	10.01	18.51	29.85	49.16	52.85	1129
Nereus full SFT	36.19	84.74	10.36	16.56	25.33	47.39	54.06	1129

Table 10: Relative changes over Nautilus counting SFT. Negative values are better for MAE and RMSE; positive values are better for all accuracy and F1 metrics.

Nereus setting	Δ MAE	Δ RMSE	Δ Acc@5	Δ Acc@10	Δ Densest Cell	Δ Camou. F1
Count-only SFT	-1.39	-68.91	+0.53	+2.13	+0.00	-1.08
QA+structured-final SFT	-7.98	-80.25	+2.21	+5.31	+43.05	+44.37
Full SFT	-3.77	-73.76	+0.27	+0.80	+41.28	+45.57

Table 11 reports the image-only ablation on InternVL2.5-8B with RGB input, isolating the effect of different counting supervision forms.

Table 11: Image-only ablation on InternVL2.5-8B with RGB input. Best values are shown in bold.

Training Setting	MAE $\downarrow$	Exact Acc. $\uparrow$	Acc@ $\pm 5\%$ $\uparrow$	Densest Cell Acc. $\uparrow$	Camouflage F1 $\uparrow$
Zero-shot	87.38	2.22	2.22	22.85	–
Prompted zero-shot	167.55	0.00	1.16	20.73	0.00
Count-only SFT	31.72	10.01	19.04	20.81	–
QA+structured-final SFT	25.99	10.45	20.46	52.70	50.02
Full Count SFT	30.48	9.74	18.95	51.90	52.59

B.2 Result Interpretation

The expanded results show that Nautilus counting supervision transfers to the Nereus test split, but mainly improves numerical counting and provides limited support for spatial localization or camouflage reasoning. Nereus count-only SFT obtains a similar MAE to Nautilus counting SFT, but its RMSE is much lower, indicating fewer severe outlier predictions.

The QA+structured-final setting is the strongest overall Track A setting for counting and spatial evidence: it achieves the lowest MAE and RMSE, the best Acc@5 and Acc@10, and a large improvement in Densest Cell accuracy. This suggests that converting point annotations into structured region and grid supervision helps the model learn where organisms are concentrated, not only how many are present. The full Nereus setting obtains the best Camouflage F1, showing that complete supervision is especially useful for visibility and camouflage-aware reasoning. The small trade-off in MAE relative to QA+structured-final SFT suggests that different supervision mixtures emphasize different parts of the task.

For Track B in Table 3, density QA supervision substantially improves density-evidence reasoning for open-source models. For example, Qwen2.5-VL-7B improves from 18.13 to 45.26 in Density F1 and from 16.04 to 37.19 in Grounding F1, while InternVL2.5-8B shows a similar gain. LLaVA-OneVision-7B reaches the best Hotspot accuracy under full SFT, but its density and grounding F1 remain lower than Qwen and InternVL. These results indicate that reading a heatmap, localizing the hotspot, and explaining the density evidence are related but not identical capabilities.

B.3 Qualitative QA Examples

Figures 8–10 show representative QA examples from the counting and density-evidence benchmark. These cases illustrate how the benchmark evaluates total counting, grid-based spatial counting, camouflage visibility, density hotspot localization, low-response interpretation, and cross-image density reasoning.

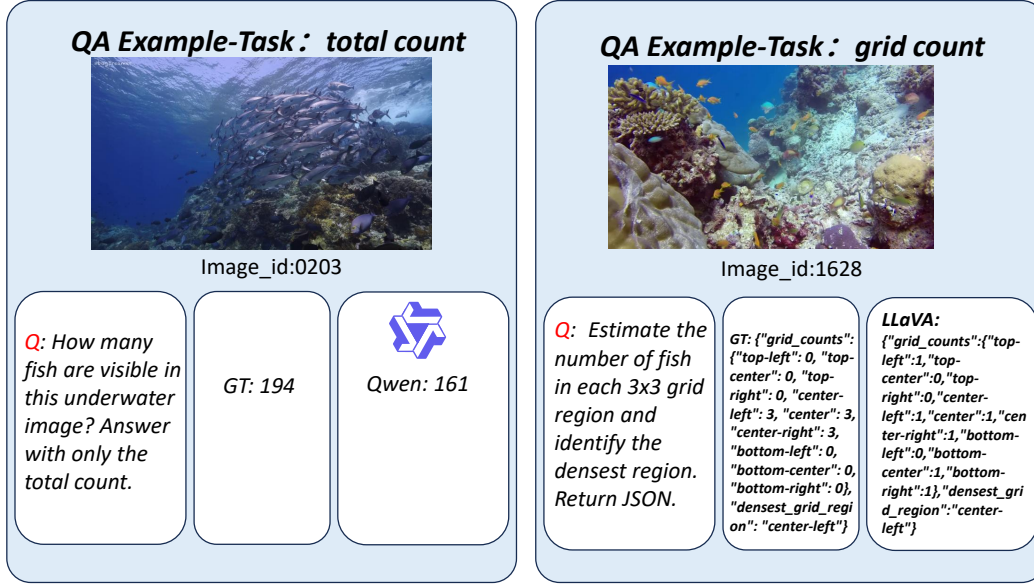


Figure 8: Qualitative examples for total-count and grid-count QA. The examples compare model outputs with grounded counting annotations, including total fish count and densest-region prediction.

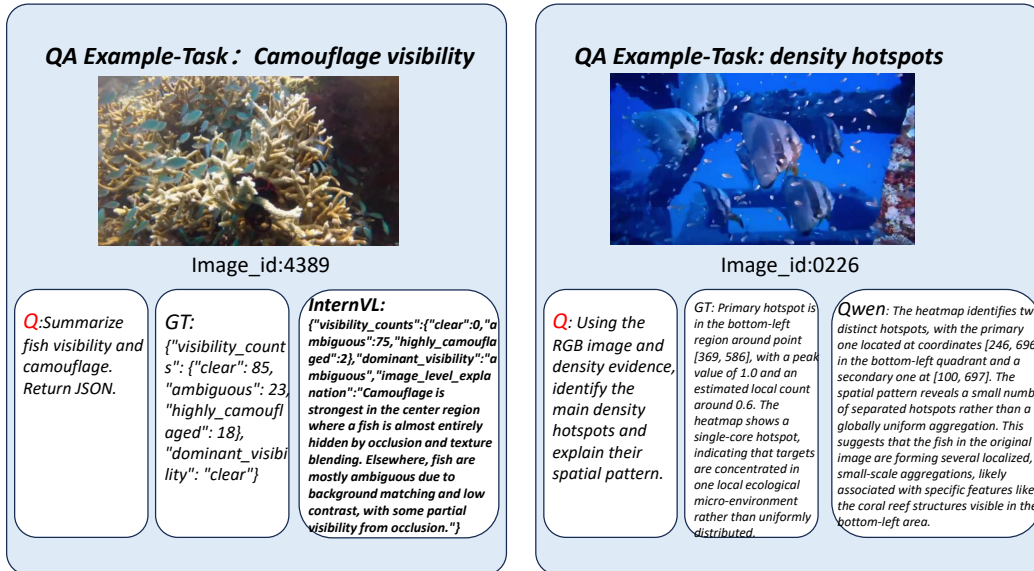


Figure 9: Qualitative examples for camouflage visibility and density-hotspot QA. The examples show visibility-state counting and hotspot reasoning from RGB images and density evidence.

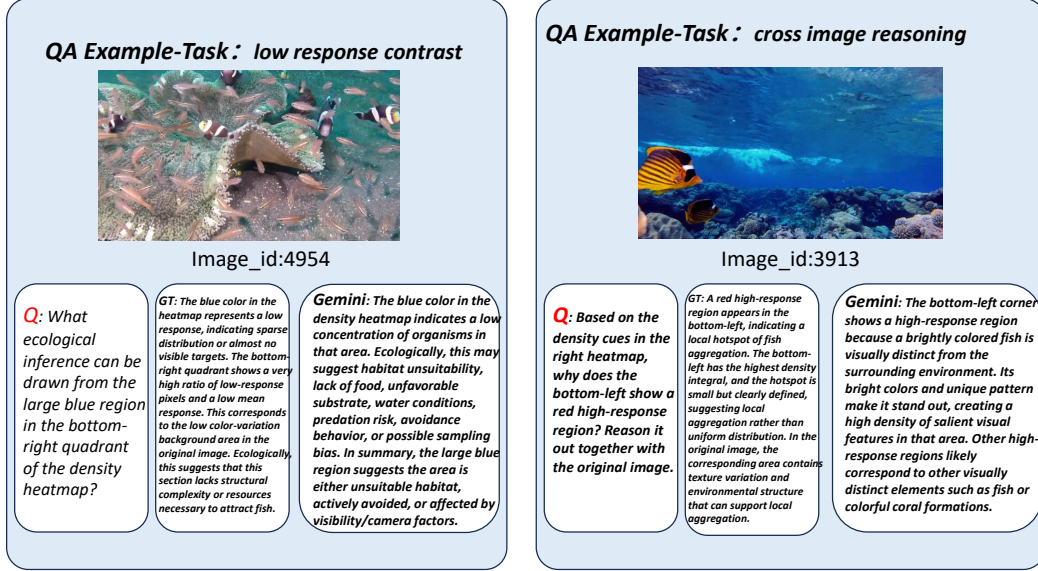


Figure 10: Qualitative examples for density-response interpretation. The examples cover ecological inference from low-response heatmap regions and cross-image reasoning over high-response density cues.

599 B.4 Metric Definitions

600 Let c_i and $\hat{c}_i$ denote the ground-truth and predicted counts for image i , and let N be the number
 601 of evaluated samples. All accuracy and F1 values are reported as percentages in the tables. Lower
 602 values are better for MAE and RMSE, while higher values are better for all other metrics.

Table 12: Metrics used in the counting and density-evidence benchmark.

Metric	Track	Computation	Interpretation
Count MAE ↓	A	$\frac{1}{N} \sum_i \hat{c}_i - c_i $	Average absolute counting error; the primary RGB counting metric.
Count RMSE ↓	A	$\sqrt{\frac{1}{N} \sum_i (\hat{c}_i - c_i)^2}$	Count error with stronger penalty for severe over-counting or under-counting.
Exact Acc ↑	A	$\frac{1}{N} \sum_i \mathbb{1}[\hat{c}_i = c_i]$	Strict exact-count accuracy; any count deviation is incorrect.
Acc@5 ↑	A	$\frac{1}{N} \sum_i \mathbb{1}[\hat{c}_i - c_i \leq 0.05c_i]$	Count accuracy under a 5% tolerance for dense scenes.
Acc@10 ↑	A	$\frac{1}{N} \sum_i \mathbb{1}[\hat{c}_i - c_i \leq 0.10c_i]$	Count accuracy under a 10% tolerance; measures count-scale correctness.
Densest Cell Acc ↑	A	$\frac{1}{N} \sum_i \mathbb{1}[\hat{r}_i = r_i^*]$	Accuracy of predicting the densest cell in a 3×3 grid; evaluates spatial counting.
Camouflage F1 ↑	A	$2PR/(P+R)$ over camouflage-related answers	F1 for visibility and camouflage reasoning over clear, ambiguous, and highly camouflaged states.
Hotspot Acc ↑	B	$\frac{1}{N} \sum_i \mathbb{1}[\hat{h}_i = h_i^*]$	Accuracy of locating the density hotspot from RGB plus heatmap evidence.
Density F1 ↑	B	F1 over normalized density-reasoning concepts	F1 over density concepts such as hotspot, low-density area, aggregation, and dispersion.
Ecology/Grounding ↑	F1 B	F1 over ecological and grounding evidence units	F1 for ecological explanations grounded in image and density evidence.

603 Overall, Track A should be read as a test of RGB-only counting, spatial localization, and camouflage
 604 understanding. Track B should be read as a test of whether a model can use explicit density evidence
 605 to localize hotspots and produce reliable grounded explanations.

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